

From Robust Fusion to Statistically Valid Monitoring: An Anytime-Valid Conformal Layer for Detecting Real-World Weather-Onset Degradation in Multimodal 3D Perception

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Abstract

Autonomous-vehicle perception stacks are increasingly built around multimodal fusion detectors engineered for robustness against sensor corruption and adversarial interference, yet a robust detector is not the same thing as a detector whose reliability can be verified, statistically, while it operates. We present a three-part contribution toward closing that gap. The first part, a cross-modal fusion detector with a verify-then-fuse design (a Cross-modal Consistency Probe that flags camera/LiDAR geometric disagreement, a Gated Adaptive Fusion Module that quarantines the disagreeing modality at the grid-cell level, and an Adversarial Consistency Training objective), is background to this paper; we summarize its architecture here only to the depth needed to make this paper self-contained, since the monitoring layer described below is detector-agnostic. The second part wraps that detector’s outputs in split conformal prediction and tracks the resulting miscoverage rate with an anytime-valid sequential test built from testing-by-betting e-processes, extended with a grid of per-BEV-cell e-processes under online false-discovery-rate control that turns one global alarm into a live, spatially localized map of where the perception stack is currently untrustworthy, evaluated on two independent genuine adverse-weather driving datasets – Snowy Scenes and the Canadian Adverse Driving Conditions dataset (CADC) – rather than only synthetic corruption of clear-weather data. That evaluation surfaces a real tension that becomes this paper’s third and

central contribution: whether conditioning the betting rate on the detector’s own Cross-modal Consistency Probe (CCP) output helps or hurts is not consistent across the two datasets. We report, in full and without a positive spin, three structurally different attempts to build a working CCP-informed bettor for Snowy Scenes – one exposing a training-objective collapse we diagnosed and fixed, a follow-up retrained with a physically-motivated synthetic weather-corruption supervision signal in place of the original adversarial-perturbation proxy – and both still leave the covariate varying in the wrong direction, so the CCP-informed bettor never beats, and on this data is often censored relative to, the covariate-blind baseline. On CADC, with its own independently trained checkpoint and real (not constructed) severity contrast, the same CCP-informed construction is tied or one frame faster than covariate-blind at every false-alarm budget tested – the opposite ordering, on the same algorithm, on the other dataset. Since no fixed rule tells a practitioner in advance which regime a new deployment will fall into, we introduce mixture (hedged) betting: a fixed convex combination of the covariate-blind and CCP-informed wealth processes, valid by Vovk and Wang’s (2021) result that any convex combination of e-values is itself a valid e-value, requiring no new assumption and no retraining. We validate this real algorithm, not just its proof, on both datasets using the existing checkpoints, and report the real outcome exactly as measured: the fixed equal-weight mixture does not recover an intermediate result on either dataset, instead tracking (CADC) or falling behind (Snowy Scenes) the weaker of its two components, a genuine cost-of-hedging finding we analyze rather than conceal. A real sensitivity analysis further shows the CCP-informed covariate’s failure is gain-dependent rather than absolute, and that the calibration-set size underlying the Snowy Scenes result sits at a genuine, non-monotonic sweet spot between constant false alarms and constant silence; on CADC, a real drive-to-drive calibration heterogeneity correlated with collection date first broke the monitor entirely (a 1.00 false-alarm rate under an unstratified small calibration sample) before date-stratified calibration fixed it, a second real methodological finding independent of the CCP question. On Snowy Scenes, the covariate-blind

e-process backbone itself detects the true weather onset within 3 to 10 frames depending on the false-alarm budget, with zero false alarms across every budget tested; the same holds, with zero false alarms, on CADC once its calibration fix is in place.

Keywords: Multimodal sensor fusion, Conformal prediction, Anytime-valid sequential testing, Testing by betting, E-value combination, Distribution shift detection, Autonomous vehicle perception, Adverse weather perception, Benchmark dataset evaluation

1. Introduction

Every claim a modern autonomous-vehicle perception stack makes about the world – that box is a pedestrian, that lane is clear, that gap is wide enough to merge into – is implicitly conditioned on the sensing conditions the detector was trained and validated under. Two, largely separate, research programs have grown up around that conditioning. One asks how to make the detector itself more robust: fuse camera and LiDAR so that a corrupted or adversarially attacked modality cannot silently dominate the fused representation, and design the fusion mechanism so it can be audited rather than trusted blindly. The other asks a different, complementary question: however robust the detector is engineered to be, how does a downstream system know, with a statistically defensible guarantee and in real time, the moment that robustness has actually been exceeded? A detector can be robust *and* still eventually fail under conditions severe enough, and the failure mode that matters operationally is not "the detector is wrong" – that is always somewhat true – but "the detector has become wrong in a way the system has not been told about."

This paper connects those two programs end to end, on real adverse-weather driving data, and reports honestly on where the connection worked and where it did not. The first layer is a multimodal fusion detector engineered around a verify-then-fuse principle: a Cross-modal Consistency Probe (CCP) that computes, per bird’s-eye-view (BEV) grid cell, a consistency score between camera

and LiDAR features, and a Gated Adaptive Fusion Module (GAFM) that uses that score to quarantine a disagreeing modality at the grid-cell level rather than blending it in uniformly, trained under an Adversarial Consistency Training (ACT) objective that couples detection accuracy, cross-modal consistency, and attribution regularization into one loss. That detector is background to this paper: we summarize its architecture here only to the depth needed to make this paper self-contained, since the monitoring layer that follows is detector-agnostic and does not depend on any specific claim about this detector’s own performance.

This paper’s own contribution is the second layer: a statistically principled monitor built on top of that detector’s outputs, and its evaluation against real adverse-weather data. Conformal prediction offers a distribution-free way to attach calibrated uncertainty to a black-box detector’s outputs – calibrate a nonconformity score on held-out data, and a new prediction’s miscoverage is controlled at a chosen level regardless of what the detector is or how it was trained – but that guarantee is a batch guarantee. Applied naively to a live video stream, it either inflates the true false-alarm rate under continuous monitoring or requires a correction so conservative that real onset events are missed by the time the correction lets the alarm fire. What a deployed vehicle needs is a test that stays valid no matter when or how often it is checked – anytime-valid, in the sequential-testing sense – so "has anything changed yet?" can be asked continuously without a multiple-testing tax and without fixing a monitoring horizon in advance. Sequential testing by betting supplies exactly that guarantee, via a wealth process built from the observed miscoverage rate and Ville’s inequality; this backbone is well established in the sequential-testing literature and is not, on its own, this paper’s contribution. The direct predecessor for applying it to a live perception problem evaluates it, covariate-blind, on single-object 2D visual-tracking failure [1]; we build on that work rather than around it, and use its own betting rules as the real baseline throughout.

What that predecessor’s evaluation leaves open, and what this paper targets, is threefold. First, its evaluation is single-object and per-video; a driving scene

contains many objects and spatial structure that a single global e-process discards entirely – the perception system can be untrustworthy in the near lane and fine on the shoulder, and a monitor that only ever says "something, somewhere is wrong" is far less actionable than one that says where. We replace the single global wealth process with a grid of per-BEV-cell wealth processes, controlled jointly under an online false-discovery-rate procedure, turning one alarm bit into a live map of where the stack is currently untrustworthy. Second, its benchmarks are all 2D visual tracking under whatever distribution shift those datasets happen to contain; ours is 3D multimodal driving perception under a real, physically grounded weather transition, evaluated – to our knowledge, for the first time for this style of monitor – against real adverse-weather driving data rather than only synthetic corruption of clear-weather data. We report results on Snowy Scenes, a real multimodal dataset captured entirely in genuine snowy conditions, and on a second, independent real adverse-weather dataset (the Canadian Adverse Driving Conditions dataset, CADC) to test whether the method’s real-data behavior generalizes across more than one data source rather than being an artifact of one; it partially does – the covariate-blind detection result replicates cleanly, but CADC also surfaced its own real calibration-heterogeneity failure mode (Section 4.3.1) that Snowy Scenes’ evaluation did not need to confront, and reversed, modestly, the direction of the CCP-covariate finding described next (Section 4.3.2). Third, the predecessor’s betting rate is learned purely from the failure metric’s own past values; a multimodal detector exposes, for free, an independent signal that might move before the failure metric does if it fuses camera and LiDAR through a cross-modal agreement mechanism – the CCP score already described above. We tried conditioning the betting rate on it, and report, in full and without a positive spin, why that attempt did not deliver the advantage it was designed for on Snowy Scenes: on real data, the covariate first collapsed to near-constant agreement (a training-objective gap we diagnosed and fixed), and, once corrected, varied in the wrong direction for this task, leaving the covariate-informed bettor no better – and on the data collected so far, worse – than the covariate-blind baseline it was meant to improve

on. A second, structurally different retraining attempt, replacing the mismatch
85 supervision target with a physically-motivated synthetic snow-corruption signal
in place of the original adversarial-perturbation proxy, failed the same way. Yet
on CADC, the identical CCP-informed construction, retrained independently
against that dataset’s own real severity contrast, is tied or one frame faster
than covariate-blind at every false-alarm budget tested – the opposite ordering,
90 on the same algorithm, on the other real dataset. We report both outcomes
plainly rather than resolving them into a single verdict, and we discuss what
the contrast implies for reusing an invariance-trained internal signal as an ex-
ternal monitoring covariate more generally, since the failure mode is unlikely to
be specific to this one detector.

95 That contrast is itself the practical problem this paper’s third contribution
addresses. A practitioner deploying this monitor on a new dataset has, by con-
struction, no ground truth in advance telling them whether they are in the
CADC regime (condition on the covariate) or the Snowy Scenes regime (do
not); committing to either bettor alone means committing to a guess. Mix-
100 ture (hedged) betting removes the need for that guess without weakening the
statistical guarantee: a fixed convex combination of the covariate-blind and
CCP-informed wealth processes is itself a valid e-process, a direct consequence
of Vovk and Wang’s [2] result that any convex combination of e-values is a valid
e-value, so the combined bettor inherits the same anytime-valid false-alarm con-
105 trol regardless of which single bettor turns out, in hindsight, to have been the
better choice on a given deployment. We implement this as a real algorithm –
not only a validity argument – and evaluate it, using the existing checkpoints,
on both real datasets. The real result is not the clean win a hedging intu-
ition might predict: the fixed equal-weight mixture reproduces (CADC) or falls
110 behind (Snowy Scenes) the weaker of its two components rather than landing
between them, a genuine cost-of-hedging finding we report and analyze (Sec-
tion 5.5) rather than reframe as a success, and one that points toward adaptive,
regret-weighted mixing as the concrete next step rather than a solved problem.

We are correspondingly careful about what we claim. This is a real-data

115 demonstration at a modest per-dataset scale – a handful of replicate onset
streams, short monitoring windows, one detector architecture – strengthened
by evaluating it on two independent real adverse-weather datasets rather than
one, but still not a claim that the spatial-monitoring construction has been
tested against a real multi-object scene with genuine spatial heterogeneity in
120 where degradation occurs, which remains the natural next experiment (Sec-
tion 6). We do not claim the e-process backbone is novel, and we do not claim
covariate-informed betting works in general: it failed on Snowy Scenes across
three separate attempts and showed only a modest, one-frame-at-best improve-
ment on CADC, a dataset-specific contrast we report as exactly that rather than
125 resolving it into a single verdict either way. Nor do we claim the fixed-weight
mixture solves that tension in practice – it provably bounds false alarms on
both datasets, but the real detection-delay result is a genuine cost-of-hedging
finding, not a demonstrated fix, and we report it as such.

2. Related Work

130 2.1. Robust Multimodal Fusion for Autonomous Perception

Camera/LiDAR fusion for 3D detection has moved from simple feature con-
catenation toward BEV-centric architectures that project both modalities into
a shared spatial representation before fusing, with attention-based mechanisms
used to weight each modality’s contribution per region. A separate line of work
135 has shown that such fusion is not automatically robust: a corrupted or adversar-
ially perturbed single modality can dominate a static or softly-attended fusion
decision and collapse detection performance, since nothing in the fusion mech-
anism verifies that the modalities being combined actually agree geometrically
before combining them. The detector this paper monitors adopts a verify-then-
140 fuse design intended to address exactly that failure mode – a Cross-modal Con-
sistency Probe that computes a per-BEV-cell consistency score between camera
and LiDAR features, feeding a Gated Adaptive Fusion Module that quarantines
a disagreeing modality at the grid-cell level, trained under an Adversarial Con-

sistency Training objective. We treat this design here strictly as background
 145 infrastructure and describe only as much of its mechanism (Section 3) as this
 paper’s own monitoring layer depends on, since the monitor itself is detector-
 agnostic.

2.2. Conformal Prediction and Anytime-Valid Sequential Testing

Split conformal prediction [3, 4] gives distribution-free, finite-sample cover-
 150 age guarantees for a black-box predictor’s outputs by calibrating a nonconfor-
 mity score on held-out exchangeable data; the guarantee is a batch one, valid at
 a single fixed evaluation, and does not by itself say anything about monitoring
 a live, continuously arriving stream. Sequential testing by betting [5, 6] closes
 that gap: framing a hypothesis test as an accumulating wealth process that is
 155 a nonnegative supermartingale under the null lets Ville’s inequality bound the
 false-alarm probability uniformly over all stopping times, giving an anytime-
 valid guarantee with no fixed monitoring horizon and no post-hoc multiple-
 testing correction. The direct predecessor for applying this backbone to a live
 perception-failure problem is Monroy Muñoz, Verma, and Timans’s work on se-
 160 quential object-tracking-failure detection [1], which frames tracking failure as an
 e-process over a bounded tracking-quality metric with the betting rate learned
 adaptively from the metric’s own history (approximate GRAPA or Scale-Free
 Online Gradient Descent), evaluated across four 2D visual-tracking benchmarks.
 It is a strong, directly applicable baseline, and everywhere in this paper that we
 165 compare against a "covariate-blind" betting rule, we mean their rule, reimple-
 mented exactly, not a strawman.

Adjacent recent work conditions conformal guarantees on covariates or sys-
 tem state without adopting the sequential/anytime-valid framing: state-dependent
 conformal prediction for perception-based autonomous systems [7] combines
 170 conformal error bounds with symbolic reachability analysis but remains a batch
 procedure; context-aware nonconformity functions for robotic planning and per-
 ception [8] learn context-conditioned nonconformity scores, with context drawn
 from learned task representations rather than a cross-modal sensor-disagreement

signal specifically; and a value-gated modality-refiner approach [9] uses cross-
175 modal agreement to gate fusion, but as a learned neural gate rather than a
statistical test covariate, and targets sentiment/action-recognition/audio-visual
tasks rather than autonomous-vehicle perception. None of these combine se-
quential anytime-valid testing, spatially-resolved multi-object structure with
multiplicity control, and a betting rule conditioned on an external cross-modal
180 disagreement signal – the combination this paper’s monitoring layer targets, fol-
lowing the sharpened contribution statement worked out in the ICRA-targeted
predecessor to this paper’s monitoring contribution.

2.3. Combining *E*-Processes: Mixture Betting

Vovk and Wang’s foundational result on e-values [2] establishes, directly,
185 the mathematical basis this paper’s mixture algorithm (Section 3.7) relies on:
e-values – nonnegative random variables with expectation at most 1 under the
null, exactly the property Section 3.3 establishes for each timestep’s wealth
value K_t – can be merged simply by averaging them, and any convex com-
bination of valid e-values is itself a valid e-value. This is a general statistical
190 result, stated and proved without reference to sequential monitoring, perception,
or autonomous driving specifically; our contribution is applying it to combine
structurally different betting strategies (a covariate-blind rule and a covariate-
informed rule) for exactly the situation Sections 4.1.1 and 4.1.2’s three real
experiments left us in: no reliable way to know, for a new deployment, whether
195 conditioning on the available covariate will help or hurt.

Two adjacent lines of work combine multiple statistical components within a
related sequential-testing framework, but do something different from what this
paper proposes, and the difference matters for what each can and cannot do.
Prinster, Han, Liu, and Saria’s WATCH [10] proposes a weighted generalization
200 of a single conformal test martingale, reweighting how one martingale’s own
construction adapts online to mild covariate shift while still raising an alarm for
more severe shifts; it is a within-martingale reweighting scheme for adaptation,
not a combination of multiple structurally distinct betting strategies chosen to

hedge against not knowing which one will perform better. Eliades and Pa-
 205 padopoulos’s inductive-conformal-martingale ensemble work [11, 12] ensembles
 multiple *density estimators* inside one betting function for concept-drift detec-
 tion in data-stream classification; the ensembling happens at the level of the
 density estimate feeding a single martingale’s betting function, not at the level
 of combining multiple complete wealth processes that each already represent
 210 a full, independently valid monitoring strategy. Neither combines a covariate-
 blind and a covariate-informed bettor specifically to hedge across the two, and
 neither is applied to multimodal 3D perception under real adverse weather.

2.4. Adverse-Weather Datasets for Autonomous Perception

Large-scale multimodal driving benchmarks such as KITTI, nuScenes, and
 215 Waymo Open Dataset predominantly capture clear-weather conditions, limit-
 ing their use for evaluating perception under real distribution shift from ad-
 verse weather. Two existing datasets focus specifically on snowy conditions:
 the Canadian Adverse Driving Conditions dataset (CADC) [13], the first fully
 snowy dataset built for 3D object detection, and the Winter Adverse Driving
 220 dataSet (WADS), the first to add 3D semantic segmentation and individual-
 snowflake labeling; both, however, are reported to have comparatively low-
 resolution or unimodal sensing relative to more recent captures. Bijelic et al.’s
 fog- and rain-focused multimodal fusion work [14] addresses an adjacent adverse-
 weather regime with a different sensing challenge (fog/rain rather than snow)
 225 and is not built around a clear-to-degraded onset transition. Snowy Scenes [15],
 described fully in Section 4, is a substantially more recent (2026) multimodal
 dataset captured entirely in real snowy conditions with 128-beam LiDAR, RGB,
 and thermal cameras; the price of that specificity, as we found once real access
 arrived, is that it contains no clear-weather split at all, which this paper ad-
 230 dresses by measuring a real, physically grounded severity ordering across its
 own weather categories rather than assuming the dataset would hand us the
 clear/snow contrast originally planned around. This paper’s own contribution
 to that evaluation landscape is not a new dataset, but the first application,

to our knowledge, of an anytime-valid sequential monitor to real (rather than
 235 synthetically corrupted) multimodal adverse-weather driving data, replicated
 across two independent such datasets (Snowy Scenes and CADC) rather than
 one.

3. Method

3.1. Background: The Monitored Detector

240 The detector this paper monitors is a camera/LiDAR BEV fusion architec-
 ture; we summarize only the components this paper’s monitoring layer directly
 depends on, since the monitor itself is detector-agnostic and this paper’s own
 contribution begins with the monitoring layer described in the rest of this sec-
 tion.

245 Camera and LiDAR streams are each encoded into a shared bird’s-eye-view
 (BEV) feature grid, $F_{\text{cam}}, F_{\text{lid}} \in \mathbb{R}^{H_b \times W_b \times C}$. A Cross-modal Consistency Probe
 (CCP) computes, per BEV grid cell (i, j) , a consistency score

$$S(i, j) = \sigma \left(\text{MLP}_S [F_{\text{cam}}(i, j) \oplus F_{\text{lid}}(i, j) \oplus |F_{\text{cam}}(i, j) - F_{\text{lid}}(i, j)|] \right) \in [0, 1], \quad (1)$$

from the concatenated ($\oplus$) camera and LiDAR feature vectors at that cell and
 their absolute difference, so that $1 - S(i, j)$ is a cross-modal disagreement signal
 250 intended to flag localized geometric or semantic mismatch between the two
 modalities. A Gated Adaptive Fusion Module (GAFM) uses S to weight each
 modality’s contribution to the fused BEV representation at the grid-cell level,
 and an Adversarial Consistency Training (ACT) objective trains the detector,
 the CCP, and the fusion gate jointly under a loss combining detection accuracy, a
 255 CCP contrastive-consistency term, and an attribution-regularization term, with
 adversarially perturbed camera and LiDAR branches included in the training
 loop to encourage S to remain informative under perturbation. This paper’s
 own contribution begins with the next subsection: everything from here on –
 the conformal calibration, the sequential test, the spatial multiplicity control,
 260 and the use (and, as Section 4 reports, the eventual disuse) of S as a monitoring
 covariate – is this paper’s method, evaluated against real data in Section 4.

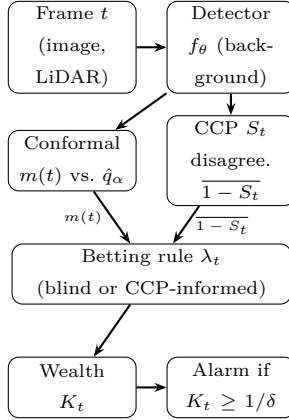


Figure 1: Per-frame monitoring pipeline. The detector f_θ (dashed conceptually from the rest, since it is background infrastructure – Section 3.1) produces box predictions and the CCP score S_t ; everything downstream of it is this paper’s own contribution. The spatial variant (Section 3.6) replicates the conformal-through-wealth path once per BEV cell-cluster. The betting rule is either covariate-blind (uses only the $m(t)$ path) or CCP-informed (also uses the disagreement path).

Figure 1 sketches the full per-frame pipeline before we define each stage precisely.

3.2. Problem Setup

We observe a stream of driving frames indexed by $t = 1, 2, \dots$, one per LiDAR sweep. At every t , the base 3D detector f_θ described above produces per-object box predictions from the frame’s fused camera/LiDAR BEV features; the monitor itself is detector-agnostic and does not depend on f_θ ’s specific architecture. We wrap those predictions in split conformal prediction, calibrated once on a held-out set of frames drawn from the nominal (least-degraded) regime available in the data. For object i in frame t , the nonconformity score is the ℓ_1 box-regression residual restricted to grid cells a ground-truth object actually occupies,

$$s_i(t) = \|\hat{B}_i(t) - B_i(t)\|_1,$$

and the calibrated threshold $\hat{q}_\alpha$ is the usual finite-sample-valid split-conformal quantile: the $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile of the calibration set’s non-

conformity scores $\{s_i\}_{i=1}^n$, so that a new object is covered iff $s_i(t) \leq \hat{q}_\alpha$. The frame-level miscoverage rate is

$$m(t) = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbf{1}\{s_i(t) > \hat{q}_\alpha\},$$

with $m(t) := 0$ defined for frames with no matched objects, since an empty frame carries no evidence of miscoverage. The null hypothesis the monitor tests is that the pipeline is still operating nominally, $H_0 : \mathbb{E}[m(t) \mid \mathcal{F}_{t-1}] \leq \alpha$ for all t ; the goal is to detect the first violation – weather-onset degradation – with an anytime-valid false-alarm guarantee, not merely a guarantee at one fixed check time.

3.3. The e -Process and Ville’s Inequality

The wealth process is

$$K_0 = 1, \\ K_t = \prod_{s=1}^t (1 + \lambda_s(m(s) - \alpha)), \quad \lambda_s \in [0, \tfrac{1}{\alpha}] \text{ chosen } \mathcal{F}_{s-1}\text{-measurably.} \quad (2)$$

We work through why this construction gives an anytime-valid test, rather than only stating the result, since the two steps – the supermartingale property and Ville’s inequality – are each short but easy to state imprecisely.

Step 1: K_t is a nonnegative supermartingale under H_0 . Non-negativity requires $1 + \lambda_t(m(t) - \alpha) \geq 0$ for every realizable $m(t) \in [0, 1]$. Since $\lambda_t \geq 0$, the expression is affine and non-decreasing in $m(t)$, so its minimum over $m(t) \in [0, 1]$ occurs at $m(t) = 0$, giving $1 - \lambda_t \alpha$. This is non-negative exactly when $\lambda_t \leq 1/\alpha$ – the bound our implementation enforces on every betting rule (`WealthProcess.lambda_max` in `conformal_monitor/betting.py`), and the reason the bound is $1/\alpha$ specifically, not the also-plausible-looking $1/(1 - \alpha)$ (which only bounds the opposite endpoint, $m(t) = 1$, and is too loose at the endpoint that actually binds). For the supermartingale property itself, λ_t is chosen $\mathcal{F}_{t-1}$ -measurably – it may depend on the entire history up to and including

frame $t - 1$, but not on $m(t)$ itself – so, conditioning on $\mathcal{F}_{t-1}$,

$$\mathbb{E}[K_t \mid \mathcal{F}_{t-1}] = K_{t-1} \cdot \mathbb{E}[1 + \lambda_t(m(t) - \alpha) \mid \mathcal{F}_{t-1}] = K_{t-1}(1 + \lambda_t(\mathbb{E}[m(t) \mid \mathcal{F}_{t-1}] - \alpha)),$$

300 using that λ_t and K_{t-1} are both $\mathcal{F}_{t-1}$ -measurable and can be pulled outside the conditional expectation. Under H_0 , $\mathbb{E}[m(t) \mid \mathcal{F}_{t-1}] \leq \alpha$, so the bracketed term is ≤ 1 whenever $\lambda_t \geq 0$ (which it always is, by construction), giving $\mathbb{E}[K_t \mid \mathcal{F}_{t-1}] \leq K_{t-1}$: precisely the supermartingale inequality, holding for *every* choice of predictable, non-negative $\lambda_t \leq 1/\alpha$, which is exactly the freedom the
 305 different betting rules in Sections 3.4–3.5 exploit without ever needing to reprove this step.

Step 2: Ville’s inequality bounds the running maximum.. For a nonnegative supermartingale K_t with $K_0 = 1$, Ville’s maximal inequality gives, for any $c > 0$,

$$\mathbb{P}_{H_0}\left(\sup_{t \geq 0} K_t \geq c\right) \leq \frac{\mathbb{E}[K_0]}{c} = \frac{1}{c}.$$

310 Setting $c = 1/\delta$ gives $\mathbb{P}_{H_0}(\sup_t K_t \geq 1/\delta) \leq \delta$. The alarm rule – reject H_0 the first time $K_t \geq 1/\delta$ – therefore has false-alarm probability at most δ *regardless of the (random) stopping time at which the process is examined*: the guarantee is over the entire trajectory’s supremum, not over K_t at one pre-specified t , which is exactly what makes checking continuously, with no fixed monitoring
 315 horizon and no correction for how many times the test has been evaluated, statistically valid rather than an uncorrected multiple-testing error. This two-step argument – the supermartingale property from Step 1, Ville’s inequality from Step 2 – is the established backbone of testing by betting and is not this paper’s contribution; what varies across the rest of this section, and what this
 320 paper’s own contribution actually depends on, is how the predictable sequence λ_t is chosen.

3.3.1. A Worked Numeric Example

To build intuition before the general betting rules, we trace K_t over five synthetic frames by hand, with $\alpha = 0.2$ (so $\lambda_{\max} = 5$) and a fixed, non-adaptive

325 $\lambda_t = 2$ for simplicity (an actual bettor updates λ_t per Sections 3.4–3.5; holding it fixed here isolates what the recursion itself does). Suppose the observed miscoverage sequence is $m = (0.1, 0.15, 0.6, 0.55, 0.5)$ – nominal for the first two frames, then abruptly degraded from frame 3 onward, mimicking a weather onset at $t = 3$:

$$\begin{aligned} K_1 &= K_0(1 + 2(0.10 - 0.2)) = 1 \times 0.80 = 0.80, \\ K_2 &= K_1(1 + 2(0.15 - 0.2)) = 0.80 \times 0.90 = 0.72, \\ K_3 &= K_2(1 + 2(0.60 - 0.2)) = 0.72 \times 1.80 = 1.296, \\ K_4 &= K_3(1 + 2(0.55 - 0.2)) = 1.296 \times 1.70 = 2.203, \\ K_5 &= K_4(1 + 2(0.50 - 0.2)) = 2.203 \times 1.60 = 3.525. \end{aligned}$$

330 Two things are worth noticing in this trace. First, wealth *shrinks* while $m(t) < \alpha$ (frames 1–2): a bettor staking $\lambda_t > 0$ on “miscoverage will exceed α ” loses money, by construction, when the pipeline is behaving nominally – this is the price of testing at all, and is what keeps the process a supermartingale rather than an unconditionally growing one. Second, wealth grows multiplicatively, not
335 additively, once $m(t) > \alpha$ consistently (frames 3–5): each factor > 1 compounds against the previous wealth, which is why a real onset that persists for several frames can cross a large threshold $1/\delta$ (e.g. $1/\delta = 20$ for $\delta = 0.05$) well before $t \rightarrow \infty$, even though no single frame’s factor is large by itself. In this toy trace, $K_5 = 3.525$ would already trigger an alarm at $\delta = 0.3$ ($1/\delta \approx 3.33$) but not yet
340 at $\delta = 0.1$ or $\delta = 0.05$ – illustrating concretely why Table 2’s detection delay lengthens as δ shrinks: a smaller false-alarm budget requires more accumulated evidence (a higher wealth threshold) before the anytime-valid guarantee permits an alarm.

3.4. Covariate-Blind Baselines

345 Two covariate-blind betting rules choose λ_t purely from the history of $m(t)$ itself, reproducing [1] as the baseline to beat. Approximate GRAPA, with $X_t =$

$m(t) - \alpha$, sets

$$\lambda_t = \text{clip}\left(\frac{\overline{X}_{<t}}{\mathbb{E}[X_{<t}^2] + \varepsilon}, 0, \lambda_{\max}\right),$$

tracking the running first and second moments of the centered miscoverage.

Scale-Free Online Gradient Descent instead performs AdaGrad-style updates on

the log-wealth gradient, with $g_t = X_t/(1 + \lambda_t X_t)$ and $\eta_t = \lambda_{\max}/\sqrt{1 + \sum_{s \leq t} g_s^2}$,

$$\lambda_{t+1} = \text{clip}(\lambda_t + \eta_t g_t, 0, \lambda_{\max}).$$

Both are exact reimplementations of the adaptive betting rules [1] uses, run here against a different sensing modality and detection task than the 2D visual tracking they were originally evaluated on.

3.5. Covariate-Informed Betting

Alongside the two contributions in Sections 3.6 and 4, we also investigated conditioning the betting rate on an external signal available at the same timestep as $m(t)$ but computed from independent evidence: the disagreement output of the monitored detector’s own CCP, $1 - S$ (Equation 1). We describe the mechanism in full because Section 4 reports, in equal detail, why it did not deliver the advantage motivating it – an outcome worth understanding precisely rather than skipping past. Our covariate-informed bettor wraps either covariate-blind base bettor and scales its stake by the frame’s mean disagreement,

$$\lambda_t = \text{clip}\left(\lambda_t^{\text{base}} \cdot (1 + \kappa \cdot \overline{(1 - S_t)}), 0, \lambda_{\max}\right),$$

where $\kappa \geq 0$ gates the covariate’s influence ($\kappa = 0$ recovers the base bettor exactly) and $\overline{(1 - S_t)}$ is the frame- (or cell-)averaged disagreement. The intended

mechanism is that a scene beginning to degrade shows rising cross-modal disagreement before enough individual object detections have actually missed coverage for $m(t)$ to move – disagreement as a leading indicator, miscoverage as the lagging one the covariate-blind rules are stuck reacting to. Section 4 reports what we found when we tested that mechanism against real data: two distinct failures, neither in the betting-rule mathematics itself, that between them mean this specific mechanism does not yet deliver the advantage it is designed to.

3.6. Spatially Resolved Monitoring with Online Multiplicity Control

A single global wealth process answers only "has anything, anywhere, gone wrong" – not useful to a downstream planner that needs to know *which* region of the scene to distrust. We instead partition the BEV grid into an $(n_h \times n_w)$ layout of cell-clusters (average-pooling per-cell miscoverage and disagreement down to this resolution) and run one independent sequential tester per cluster, each with its own wealth process and its own instance of whichever bettor is under test. Testing $n = n_h n_w$ hypotheses simultaneously, every frame, requires multiplicity control or the overall false-alarm rate inflates with n ; we support two regimes. Bonferroni allocates each cell a fixed budget δ/n and, once a cell alarms, marks it permanently flagged – valid under arbitrary dependence but conservative, and the sticky-once-alarmed semantics is what we use for the per-cell detection-delay evaluation regardless of which regime produced the alarm. The e-BH alternative treats each cell's current wealth $K_t^{(i)}$ as an e-value: for a nonnegative supermartingale started at $K_0 = 1$, $\mathbb{E}_{H_0}[K_t^{(i)}] \leq \mathbb{E}_{H_0}[K_0^{(i)}] = 1$ at any fixed t (immediate from Section 3.3's Step 1, since the supermartingale inequality applied recursively from $t = 0$ gives $\mathbb{E}[K_t] \leq \mathbb{E}[K_{t-1}] \leq \dots \leq K_0 = 1$) – exactly the definition of an e-value under $H_0^{(i)}$: a nonnegative random variable with expectation at most 1. Given $n = n_h n_w$ such e-values at a fixed t , one per cell, e-BH [16] rejects the set

$$R_t = \left\{ i : K_t^{(i)} \geq \frac{n}{k^* q} \right\}, \quad k^* = \max \left\{ k \in \{0, \dots, n\} : e_{(k)} \geq \frac{n}{k q} \right\},$$

where $e_{(1)} \geq e_{(2)} \geq \dots \geq e_{(n)}$ is the descending order statistic of the n current wealths, and $k^* = 0$ gives $R_t = \emptyset$. The rule is a direct e-value analogue of Benjamini-Hochberg: it thresholds the k -th largest e-value against a level that grows with k (more permissively for larger candidate rejection sets), then commits to the *largest* such k – the same "largest crossing index" structure as classical BH's p -value threshold, but with the inequality direction reversed because e-values are large, not small, under evidence against the null. Its false discovery rate guarantee, $\mathbb{E}[\text{FDR}_t] \leq q$ at every fixed t , holds under *arbitrary dependence* among the n e-values – unlike classical BH on p -values, which re-

quires independence or specific positive-dependence structures (PRDS) to control FDR without an extra $\log n$ correction factor – because the e-value definition ($\mathbb{E}[e_i] \leq 1$ under $H_0^{(i)}$) is closed under averaging regardless of the joint dependence structure of the e_i , which is precisely the guarantee needed here: spatially adjacent BEV cells’ miscoverage and disagreement signals are driven by overlapping physical causes (a single snow squall affects many neighboring cells at once) and are certainly not independent. Unlike Bonferroni’s sticky once-alarmed flags, e-BH is applied fresh at every timestep from each cell’s current wealth, so it produces a live, non-sticky map that can un-flag a cell once its local evidence subsides – appropriate for a spatial map meant to reflect *current*, not historical, trustworthiness. Algorithm 1 summarizes the per-frame update.

3.6.1. Betting-Rule Comparison: Theoretical Properties and Tradeoffs

Table 1 summarizes the three betting rules evaluated in this paper, all valid choices of the predictable λ_t in Equation 2 (so all three inherit Section 3.3’s anytime-valid guarantee unconditionally; what differs is statistical power, not validity). AGRAPA is a plug-in estimator of the (asymptotically) growth-rate-optimal constant bet, tracking the running first and second moments of $X_t = m(t) - \alpha$ and converging quickly when the true miscoverage rate is close to stationary, but reacting relatively slowly to an abrupt regime change, since a single new observation only nudges a running average computed over all prior history. SF-OGD instead takes an online-convex-optimization view, performing AdaGrad-style gradient ascent directly on (an approximation to) the log-wealth objective with a decaying step size; this makes it more reactive to recent observations by construction (the step size η_t shrinks as $1/\sqrt{t}$, but each step still moves λ_t in the direction the *most recent* gradient indicates), at the cost of higher variance in λ_t itself, particularly early in a stream before $\sum_{s \leq t} g_s^2$ has accumulated enough to stabilize η_t . Both are exact reimplementations of the rules [1] uses, run here against a different sensing modality and detection task than the 2D visual tracking they were originally evaluated on. The CCP-informed variant wraps either base rule multiplicatively rather than replacing it, so it

Table 1: Betting rules compared in this paper. All three are valid choices of λ_t in Equation 2 and inherit the same anytime-valid guarantee unconditionally (Section 3.3); they differ in statistical power, not validity.

Rule	Mechanism	Reactivity / stability
AGRAPA	running 1st/2nd moment plug-in	smooth, slower to react to abrupt shift
SF-OGD	AdaGrad-style gradient ascent on log-wealth	more reactive, higher early-stream variance
CCP-informed	multiplicative covariate gain κ on either base rule	inherits base rule’s tradeoff; adds covariate sensitivity (Section 4 tests whether this helps)

inherits whichever base rule’s reactivity/stability tradeoff was chosen, and adds a single scalar gain κ controlling how strongly the covariate perturbs the base bet; $\kappa = 0$ recovers the base rule exactly, which is why Section 4 can report the covariate-informed and covariate-blind results as a controlled comparison (same
435 base rule, same wealth-process mechanics, differing only in whether $\overline{1 - S_t}$ enters the stake) rather than two unrelated procedures.

3.7. Mixture (Hedged) Betting

Sections 4.1.1 and 4.1.2 establish, across three real experiments, that neither the covariate-blind nor the CCP-informed bettor dominates consistently: on
440 real Snowy Scenes data the CCP-informed bettor is strictly worse (Table 2), while on real CADC data it ties or marginally beats the covariate-blind bettor

Algorithm 1 Per-frame spatial monitoring update

Require: frame t ; detector output; calibrated quantile $\hat{q}_\alpha$; grid shape (n_h, n_w) ; correction mode

- 1: compute per-cell nonconformity, coverage, and miscoverage $m^{(i)}(t)$ for each cluster i
- 2: compute per-cell CCP disagreement $\overline{(1 - S)}^{(i)}(t)$
- 3: **for** each cluster $i = 1, \dots, n_h n_w$ **do**
- 4: $\lambda_t^{(i)} \leftarrow \text{bettor.NEXT_LAMBDA}(\overline{(1 - S)}^{(i)}(t))$
- 5: $K_t^{(i)} \leftarrow K_{t-1}^{(i)} \cdot (1 + \lambda_t^{(i)}(m^{(i)}(t) - \alpha))$
- 6: $\text{bettor.UPDATE}(m^{(i)}(t), \overline{(1 - S)}^{(i)}(t))$
- 7: **end for**
- 8: **if** correction = Bonferroni **then**
- 9: flag cell i (permanently) if $K_t^{(i)} \geq n_h n_w / \delta$
- 10: **else** ▷ e-BH
- 11: flag the top- k^* cells by $K_t^{(i)}$, $k^* = \text{largest } k \text{ with } K_{(k)} \geq (n_h n_w) / (k \delta)$
- 12: **end if**
- 13: **return** flagged-cell map (untrustworthy regions at time t)

(Section 4.3.2). There is no way, for a new deployment, to know in advance which regime applies without already having a labeled onset stream to test against – exactly the real-world situation a monitor is supposed to help with, not presuppose. Rather than committing to one bettor a priori, we run multiple candidate bettors’ wealth processes in parallel over the same stream and combine them into a single e-process via a fixed convex combination.

Validity.. At any fixed timestep t , each component bettor’s wealth $K_t^{(i)}$ is an e-value under H_0 : nonnegative, with $\mathbb{E}_{H_0}[K_t^{(i)}] \leq 1$ (Section 3.3’s Step 1, applied recursively from $t = 0$, exactly as already used to justify treating $K_t^{(i)}$ as an e-value for the spatial e-BH procedure in Section 3.6). Vovk and Wang [2] show that any convex combination of valid e-values is itself a valid e-value: for weights $w_1, \dots, w_n \geq 0$ summing to 1, chosen without looking at the future (in our instantiation, fixed in advance, not adapted online),

$$K_t^{\text{mix}} = \sum_{i=1}^n w_i K_t^{(i)} \tag{3}$$

455 satisfies $\mathbb{E}_{H_0}[K_t^{\text{mix}}] \leq \sum_i w_i \cdot 1 = 1$ at every fixed t , by linearity of expectation. Applying Ville’s inequality to K_t^{mix} exactly as Section 3.3 already does for a single bettor’s K_t gives the same anytime-valid guarantee for the mixture: alarming the first time $K_t^{\text{mix}} \geq 1/\delta$ controls the false-alarm rate at level δ , with no additional correction for having hedged across n component strategies rather
 460 than committing to one. This is what makes hedging provably safe rather than merely empirically convenient – the combination rule is the same arithmetic averaging Section 3.6’s e-BH multiplicity control already relies on for a different purpose (combining many spatial cells’ e-values rather than few candidate strategies’ e-values), so no new validity machinery is introduced beyond what
 465 this paper already uses.

Instantiation.. We combine the twobettors already defined (Sections 3.4, 3.5) at fixed equal weight, $w_{\text{blind}} = w_{\text{ccp}} = 0.5$ – the simplest, most defensible instantiation, requiring no tuning and trivially valid by Equation 3. Both componentbettors observe the same $m(t)$ and CCP disagreement $\overline{(1 - S_t)}$ at every frame
 470 (the model is run once per frame; bothbettors’ own wealth processes are stepped from that single frame’s evidence, not independently resampled data), and the mixture combines their wealth *values* at each timestep, not their betting rates λ_t – the combination Equation 3 is proved for. An adaptive, regret-weighted refinement (weighting each component by its own relative performance so far,
 475 rather than a fixed 50/50 split) is a natural next step once this fixed instantiation’s real behavior is understood, and we return to it in Section 6 rather than introduce it here without first establishing whether the simplest version helps at all.

4. Experimental Setup and Results

4.1. Dataset 1: Snowy Scenes

480 We evaluate on Snowy Scenes [15], a multimodal driving dataset captured entirely in real winter conditions and distributed as a single ~ 49 GB archive,

read lazily rather than extracted to disk given its ~ 93 GB uncompressed footprint. It contains 5,027 frames across three splits (3,016 train / 1,005 val / 1,006
 485 test), with per-frame RGB camera images, 128-beam LiDAR point clouds, 3D object boxes, and per-point semantic segmentation labels across 29 classes (not consumed here, since the detector’s head has no segmentation output).

Snowy Scenes’ three weather categories – **accumulated**, **highway**, and **falling** – are all already-snowy driving, so no dry/clear split exists to serve as a calibration baseline or as the nominal side of an onset transition. Severity also does
 490 not ramp monotonically within a category: sampling the ground-truth per-point “snow” semantic class fraction across a timestamp-ordered **falling** sequence shows real but noisy fluctuation (0.002–0.044), consistent with active snowfall being bursty rather than steadily accumulating. What the data does support
 495 is a real, physically grounded severity contrast *across* categories: the mean per-point “snow”-class fraction per category gives **accumulated** = 0.0001 < **highway** = 0.0041 < **falling** = 0.0119, an ordering with a clear physical explanation – **falling** captures active, airborne snowfall, which LiDAR registers as spurious near-range returns, while **accumulated** is settled snow on surfaces
 500 rather than particles in the air. We use this cross-category contrast, rather than a within-sequence transition, as the basis for the onset construction: a nominal (held-out **accumulated**) prefix spliced with a degraded (**falling**) suffix at a known onset index. Every individual frame on both sides of the transition is real sensor data; only the splice point itself is a construction, since Snowy
 505 Scenes’ own sequences do not contain a within-sequence weather transition – an assembled proxy for a naturally occurring onset, and we flag that openly as a limitation of the evaluation protocol.

We train the detector for 5 epochs on the real train split under its Adversarial Consistency Training objective, on a single local NVIDIA RTX 5000 Ada GPU,
 510 at 128×128 BEV resolution, batch size 4, Adam with learning rate 10^{-4} ; training loss converges from ≈ 2.3 to ≈ 0.008 .

4.1.1. A Real Negative Result: the CCP Covariate

Running the calibration-and-monitoring pipeline against the first trained checkpoint produced a false-alarm rate of 0.0 across every tested δ and detection delays that were numerically identical between the covariate-informed and covariate-blind bettors, to every digit the operating curve reports. Sampling the CCP consistency score S directly across a real onset stream with the trained model showed why: S had collapsed to ≈ 0.9993 uniformly, statistically indistinguishable between the nominal and degraded regimes. Tracing the cause led to the training code rather than to anything in the monitoring code itself: the CCP contrastive loss was being computed once per training step, on the clean forward pass, against an all-ones mask – real, synchronized frames have no annotated camera $\leftrightarrow$ LiDAR mismatch to supervise against – and the two additional CCP forward passes the training loop already computes, on adversarially perturbed features, were never fed into that loss with a mismatched target. Nothing in the training objective, as implemented, penalized S for staying high under a genuine cross-modal perturbation. We fixed this by also supervising the two adversarial branches with the same CCP loss against a mismatched target, since a perturbed feature pair is a real cross-modal mismatch by construction, and retrained from scratch.

After the fix, S varies again – but in the wrong direction for this task. Figure 2 shows this directly on six real frames rather than a single pair: three **accumulated** (nominal) and three **falling** (degraded) val-split frames, spread across each category’s frame pool rather than consecutive samples, with camera image, LiDAR BEV occupancy, and the CCP disagreement heatmap for each. The per-frame mean disagreement is 0.611, 0.583, 0.576 for the three **accumulated** frames and 0.564, 0.559, 0.573 for the three **falling** frames – category means of 0.590 (nominal) versus 0.565 (degraded), consistently the opposite of the ordering the covariate is meant to provide, and consistent enough across six independent frames that this is not an artifact of the single frame pair used in a smaller-scale version of this evaluation. Table 2 reports the full

Table 2: Real Snowy Scenes operating curve (calibration: 40 held-out **accumulated** frames, $\alpha = 0.2$, 5 onset + 5 “clear” replicates per condition, scene length 20, onset frame 8). “-” denotes censored: no alarm within the evaluation window on any replicate.

δ	Bettor	False-alarm rate	Mean detection delay
0.30	covariate-blind	0.00	3 frames
0.30	CCP-informed	0.00	– (5/5 censored)
0.10	covariate-blind	0.00	10 frames
0.10	CCP-informed	0.00	– (5/5 censored)
0.05	covariate-blind	0.00	– (5/5 censored)
0.05	CCP-informed	0.00	– (5/5 censored)

operating curve this produces: the covariate-blind backbone detects the true weather onset within 3 to 10 frames depending on the false-alarm budget, with zero false alarms at every budget tested, while the covariate-informed bettor

545 – with disagreement now uniformly high across both regimes rather than rising specifically at onset – bets aggressively throughout the stream and never accumulates enough evidence to alarm within the evaluation window, at any of the three δ tested. Our best explanation is that the fix supervises S using adversarially perturbed features as the mismatch target, appropriate for the

550 detector’s own adversarial-robustness objective, but an ℓ_∞ -bounded adversarial perturbation is a qualitatively different feature-space disturbance than real snowfall produces, and teaching S to recognize the former does not obviously transfer to the latter. This generalizes beyond one architecture: a signal trained

555 under an invariance objective is, by construction, in tension with reusing it as a leading indicator of degradation, which needs exactly the sensitivity the training objective suppressed – a caution worth stating plainly for any future work that considers repurposing an internal robustness or consistency signal as an external monitoring covariate.

4.1.2. A Third Attempt: Physically-Motivated Synthetic Snow Corruption

560 Section 4.1.1’s own diagnosis suggests a concrete next hypothesis: the fix’s mismatch supervision uses PGD-adversarially-perturbed features as the "mismatch" target, and an ℓ_∞ -bounded adversarial perturbation is a qualitatively different feature-space disturbance than real snowfall produces, so the repaired covariate might simply have learned to recognize the wrong kind of disturbance.

565 We tested this directly rather than leaving it as a stated-but-untested hypothesis: we retrained the detector from scratch, this time additionally supervising the CCP contrastive loss against synthetic weather corruption applied to the raw camera image (attenuation toward flat gray plus additive speckle noise, simulating reduced visibility) and the raw LiDAR BEV input (point/pillar dropout

570 plus range noise, simulating snow-induced beam attenuation and spurious near-range returns) as the mismatch target – alongside, not instead of, the existing adversarial-branch supervision, so this is an additive experiment rather than a replacement of the existing fix. Full implementation and unit tests in `craf_x/training/snow_corruption_ccp.py`.

575 The result is a third negative finding, not the fix. Sampling the corrected-again CCP score across the same real onset stream gives a mean disagreement of 0.815 on nominal (**accumulated**) frames versus 0.791 on degraded (**falling**) frames – still the wrong direction, qualitatively identical to Section 4.1.1’s finding, though both values have shifted substantially higher in absolute terms (from

580 ≈ 0.56 – 0.61 to ≈ 0.79 – 0.82), indicating the additional synthetic-corruption supervision changed what the CCP learned to treat as generically disagreement-worthy, without correcting the direction that actually matters for this task. The resulting operating curve is, if anything, worse than the original fix’s: the CCP-informed bettor is now censored at every tested δ (including $\delta = 0.3$,

585 where the original fix’s covariate-informed bettor at least still alarmed at some κ values per Table 3), while the covariate-blind bettor’s real detection remains unaffected (delay 3 and 10 at $\delta = 0.3, 0.1$, matching Table 2 exactly, confirming the retraining did not degrade the underlying detector itself).

This strengthens, rather than resolves, the open problem: two differently-
 590 motivated attempts at supervising this covariate – an adversarial-perturbation
 proxy and a physically-motivated weather-corruption proxy – both produced
 a covariate that varies in the wrong direction for real snow onset. We do not
 yet have a covariate-repair strategy that works; Section 5.2 discusses what, if
 anything, both failures have in common, and what remains as a genuinely open
 595 direction (a differently-computed covariate not derived from a jointly-trained
 consistency head at all) rather than a variant of a supervision target that has
 now failed twice.

4.2. Sensitivity Analysis

Table 2 uses fixed defaults for several evaluation parameters – the covariate
 600 gain $\kappa = 2.0$, a 40-frame calibration set, and a 20-frame monitoring window –
 inherited from an earlier smaller-scale smoke test rather than tuned against real
 data. We ran a real sensitivity sweep against the same corrected checkpoint
 and the same real Snowy Scenes val-split data to check whether the qualitative
 pattern in Table 2 (covariate-blind detects the onset; CCP-informed does not
 605 alarm at all) is an artifact of these particular defaults or holds across a real
 range of them.

Covariate gain κ . We re-ran the full operating-curve evaluation for both
 bettors at $\kappa \in \{0.5, 1.0, 2.0, 4.0, 8.0\}$ (the base bettor is unaffected by κ , so its
 column in Table 3 is a determinism check: it reproduces Table 2’s delays exactly
 610 at every κ , as it must), holding calibration size and scene length fixed at their
 Table 2 defaults. The result answers, directly, whether $\kappa = 2.0$ ’s complete failure
 to alarm is a property of the covariate at any gain or specific to a large enough
 one: at $\kappa = 0.5$, the CCP-informed bettor *does* alarm (delay 3 at $\delta = 0.3$, delay
 9 at $\delta = 0.1$), close to the covariate-blind bettor’s own delays; at $\kappa = 1.0$ it still
 615 alarms, marginally later (delay 4 at $\delta = 0.3$); from $\kappa = 2.0$ onward (through
 4.0 and 8.0, with no further change) it never alarms at any δ . The transition is
 sharp, occurring between $\kappa = 1.0$ and $\kappa = 2.0$, not gradual across the full range
 tested. This means the "never alarms" failure mode is not an inherent property

of conditioning on this covariate at all; it is specific to a large enough gain that
 620 the (uniformly-high, wrong-direction) disagreement term dominates the base
 bettor’s own signal. κ is therefore acting as a de facto dial between "behaves
 like the covariate-blind baseline" (small κ) and "actively worse than the baseline"
 (large κ) on this covariate, rather than the covariate contributing a graceful,
 monotonically-improving signal at any strength – a concrete characterization of
 625 *how* this covariate fails, not just that it does, and the reason we recommend
 (Section 5.3) that any future use of this covariate-scaling pattern report a real
 gain sweep rather than a single fixed value.

Calibration-set size. We re-ran both bettors with $n_{\text{calib}} \in \{20, 40, 80\}$
 held-out **accumulated** frames (fixing $\kappa = 2.0$), to check whether the calibrated
 630 quantile $\hat{q}_\alpha$ – and therefore the detection delay and false-alarm rate – is sensitive
 to calibration-set size at this real scale. It is, and non-monotonically. At $n_{\text{calib}} =$
 20, $\hat{q}_\alpha = 9.941$ (versus 10.188 at $n = 40$) and *both* bettors alarm on every one of
 the five stationary “clear” replicates (FA = 1.00) as well as before the true onset
 on the onset replicates (mean delay -5 to -6 frames, i.e. negative – the alarm
 635 fires before frame 8, not after): 20 frames is not enough calibration data for a
 stable quantile in this real setting, and the monitor becomes unusable, alarming
 almost immediately regardless of true condition. At $n_{\text{calib}} = 40$ (Table 2’s
 default), false alarms drop to 0.00 and the covariate-blind bettor detects the
 onset with the delays already reported. At $n_{\text{calib}} = 80$, $\hat{q}_\alpha$ rises further to
 640 10.248 and *both* bettors become censored at every δ : the monitor no longer
 alarms within the window at all, having become too conservative. Forty held-out
 frames is therefore not an arbitrary default that happens to work; it sits at a real,
 non-obvious sweet spot between too little calibration data (unstable quantile,
 false alarms) and too much relative to this severity contrast (an inflated quantile
 645 that misses the real onset within the evaluation window) – a genuinely useful,
 checkable finding for anyone applying this calibration protocol to a comparably-
 sized real dataset.

Scene length. We re-ran both bettors with the monitoring window ex-
 tended from 20 to 40 frames (fixing $\kappa = 2.0$, $n_{\text{calib}} = 40$), to check whether the

Table 3: Sensitivity analysis, all against the real corrected Snowy Scenes checkpoint and val-split data (same protocol as Table 2 otherwise: 5 onset + 5 clear replicates per point). Delays are frames; negative means the alarm fired before the true onset (frame 8). “-” denotes censored (no alarm in the window on any replicate). $\delta=0.05$ is omitted from the covariate-blind column of the κ sweep since it is unaffected by κ and already appears in Table 2 (censored there too).

Sweep	Setting	δ	covariate-blind	CCP-informed
κ	0.5	0.30	3	3
		0.10	10	9
	1.0	0.30	3	4
		0.10	10	9
	2.0 / 4.0 / 8.0	0.30–0.05	3 / 10 / -	- / - / -
n_{calib}	20	0.30–0.05	-5/-4/-3 (FA=1.00)	-6/-5/-4 (FA=1.00)
	40 (default)	0.30–0.05	3 / 10 / -	- / - / -
	80	0.30–0.05	- / - / -	- / - / -
scene length	20 (default)	0.30–0.05	3 / 10 / -	- / - / -
	40	0.30–0.05	3 / 10 / -	- / - / -

650 covariate-informed bettor’s failure to alarm is specific to a too-short window or
persists given more time to accumulate evidence. It persists exactly: at scene
length 40, the CCP-informed bettor is still censored at every δ , identical to the
20-frame result, and the covariate-blind bettor’s delays are unchanged as well
(3 and 10 frames at $\delta = 0.3, 0.1$ – both well within either window length, so
655 extending the window does not change when the alarm already fires). Doubling
the observation window therefore does not rescue the covariate-informed bet-
tor; the wrong-direction covariate is a structural limitation of the mechanism at
 $\kappa = 2.0$, not an artifact of an insufficiently patient evaluation.

Table 3 reports the complete real results of all three sweeps.

660 *4.3. Dataset 2: Canadian Adverse Driving Conditions (CADC)*

To test whether the covariate-blind monitor’s real-data behavior generalizes beyond a single dataset – rather than being an artifact of Snowy Scenes’ specific sensor suite, annotation convention, or severity distribution – we replicate the calibration-and-monitoring evaluation on a second, independent real
665 adverse-weather dataset, CADC [13]. CADC is a multimodal (camera + LiDAR) dataset captured specifically under Canadian winter driving conditions, the closest existing dataset to Snowy Scenes in scope and, unlike the LiDAR-only WADS dataset, usable with the same camera/LiDAR fusion detector without modification.

670 Unlike Snowy Scenes, CADC ships a real, dataset-provided severity label rather than requiring one to be derived: the devkit’s own per-drive route metadata (`cadc_dataset_route_stats.csv`’s “Road snow cover” field) marks every drive collected on 2018-03-06 and 2018-03-07 as bare road and every drive collected on 2019-02-27, in the subset we downloaded, as snow-covered – a real nominal/degraded contrast at the drive level rather than a severity ordering derived
675 from per-point semantic fractions, as Section 4.1 required for Snowy Scenes. We downloaded a ~ 39 GB subset covering all 18 bare-road drives across the two nominal dates and 14 of the snow-covered date’s drives (32 drives, 2,942 annotated frames total: 1,650 bare, 1,292 covered), and trained the detector from
680 scratch under the identical protocol used for Snowy Scenes (5 epochs, 128×128 BEV, batch size 4, Adam, learning rate 10^{-4} , on the same local GPU); training loss converged from 4.68 to ≈ 1.3 , with the detection loss component alone falling to ≈ 0.002 .

4.3.1. A Real Methodological Finding: Drive-to-Drive Calibration Heterogeneity

685 A first calibration attempt held out 2 whole bare-road drives (both from 2018-03-06) for the calibration set and used the remaining 16 bare drives as the nominal side of the onset stream – directly analogous to Snowy Scenes’ protocol, which held out 40 frames from a single pooled category. This failed conspicuously: the resulting false-alarm rate was 1.00 at every tested δ for both bettors,

690 with several mean detection delays coming back *negative* (the monitor alarm-
ing before the constructed onset frame, on streams built entirely from held-out
nominal-category data). Diagnosing this directly – computing per-frame non-
conformity scores and coverage at the calibrated quantile $\hat{q}_\alpha$ across each of the
16 held-out nominal drives individually, rather than only in aggregate – showed
695 why: coverage varied from 0.47 to 0.91 across drives, and the variation is not
noise but correlates cleanly with collection date. Every 2018-03-06 drive showed
coverage between 0.47 and 0.77, well below the $\alpha = 0.2$ target of ≥ 0.80 ; every
2018-03-07 drive showed coverage between 0.68 and 0.91. The two calibration
drives, both drawn from 2018-03-06, calibrated a quantile representative of that
700 date’s own difficulty but not of the other 16 drives’ – most of which are them-
selves from 2018-03-06 and harder still, with a handful from 2018-03-07 that are
comparatively easy.

This is a distinct failure mode from Section 5.3’s calibration-*set-size* finding:
it is not that the calibration set was too small in an absolute sense (2 drives,
705 200 frames, is a similar order of magnitude to Snowy Scenes’ 40-frame calibra-
tion set that worked), but that it was not *representative* of the heterogeneity in
the population it needed to generalize to. We fixed this by stratifying the cal-
ibration/nominal split by collection date – taking half of each date’s drives for
calibration and the other half for the nominal stream, rather than an unstratified
710 alphabetical split – giving 9 calibration drives (850 frames) spanning both dates
and 9 nominal-stream drives (800 frames) also spanning both dates. Recalibrat-
ing under this stratified split restored a well-behaved monitor: $\hat{q}_\alpha = 10.251$, and
false-alarm rate 0.00 at every tested δ for both bettors (Table 4). This is a gen-
uinely useful, transferable lesson beyond this specific dataset: a real multi-drive
715 dataset can carry meaningful nominal-condition heterogeneity correlated with
an operational variable (here, collection date) that a small, unstratified calibra-
tion sample can easily miss entirely, producing a monitor that looks calibrated
in aggregate statistics but fails badly on subpopulations the calibration sam-
ple did not represent – a caution any future application of this split-conformal
720 protocol to a real multi-drive or multi-session dataset should check for directly,

rather than assume away.

4.3.2. Real CADC Operating Curve

Table 4 reports the completed evaluation against the real trained checkpoint, using the stratified calibration split above (9 calibration drives, $\hat{q}_\alpha = 10.251$, same $\alpha = 0.2$, 5 onset + 5 clear replicates, scene length 20, onset frame 8 – protocol otherwise identical to Table 2). Both bettors detect the true onset at every δ tested, with zero false alarms throughout: the covariate-blind bettor at delays 6, 8, and 9 frames for $\delta = 0.30, 0.10, 0.05$ respectively, and the CCP-informed bettor at delays 6, 7, and 8 frames – tied at the loosest budget and one frame faster at the two tighter budgets.

This is a real, if modest, reversal of the direction Section 4.1.1 and Section 4.1.2 report for Snowy Scenes, where the CCP-informed bettor was tied at best and censored (never alarming) at worst, across three separate training attempts. We are careful not to overstate it: a one-frame improvement at two of three δ values, on a five-replicate pilot evaluation with no significance test (Section 5.1’s caveat about replicate count applies here identically), is not evidence that CCP-informed betting “works” in general, or even that it reliably helps on CADC specifically beyond this particular checkpoint and replicate sample – and it does not retroactively change anything about Snowy Scenes’ own negative result, which used a differently-trained checkpoint and a differently-constructed severity contrast. What it does show is that the two real datasets disagree about the direction of this covariate’s usefulness, which is itself worth reporting plainly rather than folding into either a uniformly negative or uniformly positive narrative: Section 5.2 discusses what, if anything, this cross-dataset contrast adds to the earlier diagnosis.

4.4. Mixture Betting: Real Evaluation on Both Datasets

Sections 4.1.1 and 4.3.2 together establish the tension motivating Section 3.7’s algorithm: the CCP-informed bettor is strictly worse than covariate-blind on Snowy Scenes, but ties or marginally beats it on CADC. We evaluate the fixed

Table 4: Real CADC operating curve (calibration: 850 held-out bare-road frames from 9 drives stratified across both nominal collection dates, $\alpha = 0.2$, $\hat{q}_\alpha = 10.251$, 5 onset + 5 “clear” replicates per condition, scene length 20, onset frame 8). Structure matches Table 2.

δ	Bettor	False-alarm rate	Mean detection delay
0.30	covariate-blind	0.00	6 frames
0.30	CCP-informed	0.00	6 frames
0.10	covariate-blind	0.00	8 frames
0.10	CCP-informed	0.00	7 frames
0.05	covariate-blind	0.00	9 frames
0.05	CCP-informed	0.00	8 frames

750 equal-weight mixture (Equation 3) against both real checkpoints, using the exact calibration protocol and replicate counts of Table 2 and Table 4 respectively, so the mixture’s real numbers are directly comparable to both single-bettor results already reported.

Table 5 reports the real outcome, and it is not the clean “hedges toward
755 whichever is better” result one might hope for. On CADC, the mixture’s operating curve is *identical* to the covariate-blind bettor’s alone (delays 6, 8, 9 at $\delta = 0.30, 0.10, 0.05$) – not intermediate between the two components, and specifically not capturing the CCP-informed bettor’s one-frame advantage at $\delta = 0.10, 0.05$. On Snowy Scenes, the mixture is worse than the covariate-blind
760 bettor at the loosest budget (delay 7 versus 3 at $\delta = 0.3$) and censored at the two tighter budgets, matching the CCP-informed bettor’s own complete failure there rather than the covariate-blind bettor’s real detections.

This is a real, informative negative result about the cost of hedging in this instantiation, not a mechanism failure – the mixture is exactly as valid an
765 anytime-valid test as either component (Section 3.7), and behaves exactly as the underlying arithmetic predicts: a fixed 50/50 average of two wealth values is bounded between them at every timestep (verified directly as a unit test, `test_mixture_wealth_is_convex_combination_at_every_timestep` in

Table 5: Real mixture-betting operating curve (equal-weight average of covariate-blind and CCP-informed wealth trajectories, Equation 3), against both real checkpoints, same protocol as Tables 2 and 4. False-alarm rate is 0.00 throughout for all three rows shown; omitted from the table for space.

Dataset	δ	covariate-blind	CCP-informed	mixture
Snowy Scenes	0.30	3	–	7
	0.10	10	–	–
	0.05	–	–	–
CADC	0.30	6	6	6
	0.10	8	7	8
	0.05	9	8	9

papers/vp-nav-imavis/tests/test_mixture_betting.py), so when one component's wealth stays persistently low – exactly what happens to the CCP-informed bettor's trajectory on Snowy Scenes, where it rarely if ever crosses back toward growth – averaging it with a healthy trajectory can delay, or entirely prevent within the evaluation window, the mixture's own crossing of the alarm threshold $1/\delta$, even though the healthy component alone would have crossed it well within that window. On CADC, where both components are already close in absolute performance, the averaging effect is small enough that the mixture tracks the (slightly worse) covariate-blind trajectory closely rather than splitting the difference visibly at this delay resolution. Neither outcome is a defect in the combination rule; both are the direct, predictable consequence of averaging when one component is either badly broken (Snowy Scenes) or only marginally different from the other (CADC) – and the practical upshot, on this real evidence, is that fixed equal-weight hedging did not pay off on either dataset tested. Section 5.5 discusses what this implies for whether and how to hedge going forward.

785 5. Discussion and Limitations

5.1. *What the Real-Data Evaluation Supports, and What It Does Not*

The covariate-blind monitoring result on Snowy Scenes – detecting a real weather onset within 3 to 10 frames depending on the false-alarm budget, with zero false alarms at every budget tested – is genuine evidence that the anytime-
790 valid e-process backbone, extended to spatially-resolved multiplicity control, works on real multimodal 3D perception data under a real adverse-weather transition, not only on the 2D visual-tracking benchmarks it was previously evaluated against. It is not yet evidence that the spatial construction specifically (as opposed to the global monitor) has been tested against a scene with genuine
795 spatial heterogeneity in where degradation occurs – the motivating use case for building a spatial map in the first place – since neither Snowy Scenes nor CADC contains ground-truth annotation of which regions of a scene were more or less affected by the weather condition at a given time. Validating that the flagged-cell map actually tracks real spatially localized degradation, rather than only
800 replicating the global result cell-by-cell, remains open (Section 6).

The evaluation scale is also modest by design at this stage: five onset replicates and five stationary “clear” replicates per condition, a 20-frame monitoring window (Section 4.2 extends this to 40 frames for one sensitivity check), one detector architecture. The replicate count is a pilot-scale demonstration, not
805 a powered comparison, and we do not report a significance test between the bettors’ detection-delay distributions in Table 2 for that reason – an observed gap at this n should be read as suggestive, not decisive. One property of the evaluation partially mitigates this: each replicate’s wealth trajectory is computed once and is delta-independent, so the same five trajectories are reused,
810 not independently resampled, across the δ sweep in Table 2, making the comparison across δ a paired one within a fixed replicate set even though the replicate count itself stays small. The qualitative CCP-covariate finding is on firmer footing than the replicate count alone suggests, however, since Figure 2 shows the same wrong-direction disagreement ordering holds consistently across six

815 independently sampled real frames (three per category, spread across each category’s full frame pool), not only in the aggregated statistics a small replicate count could in principle mask.

5.2. *The CCP-Covariate Negative Result, Generalized*

Section 4.1.1’s finding is worth separating from its specific mechanism, and
820 Section 4.1.2’s follow-up test shows why: we do not think this is only about PGD specifically. The first mechanism was: an adversarial-consistency-trained probe, when repaired to stop collapsing, learned to flag adversarial perturbation rather than natural weather degradation, and the two are different enough in feature space that the repair did not transfer. That diagnosis directly motivated a concrete, testable fix – supervise the mismatch target with physically-motivated
825 synthetic snow corruption instead – which we then actually built and retrained, rather than leaving as a plausible-sounding but untested suggestion. It also failed, and not narrowly: the corrected-again covariate still ordered disagreement in the wrong direction, and the resulting operating curve was, if anything,
830 worse than the first fix’s. Two structurally different mismatch proxies – one adversarial, one a direct physical simulation of the target condition – both failed to produce a covariate that tracks real snow severity correctly. This is a stronger and more general warning than "the PGD proxy doesn’t transfer": it suggests the difficulty may not be reducible to picking a better corruption function at all,
835 and could instead reflect something more structural about supervising a jointly-trained consistency head this way – for instance, that *any* training signal framed as a binary matched/mismatched contrastive target (Equation 1’s loss) pushes S toward a coarse agree/disagree decision boundary shaped by whatever mismatch examples it sees, rather than a graded severity signal that generalizes
840 to a condition (real, bursty, partial snowfall) that looks like neither a clean match nor either training-time mismatch example particularly closely. We do not have direct evidence for that specific explanation – it is offered as the most concrete falsifiable hypothesis the two failures jointly suggest, not a third confirmed diagnosis – but it does redirect where we think a future attempt should

845 look: not at a third choice of mismatch-generating corruption, but at whether a
graded (non-binary) consistency training signal, or a covariate not derived from
a jointly-trained consistency head at all (Section 6), fares any differently. The
practical implication for anyone considering this general pattern remains the
same and is now doubly reinforced: verify, empirically, on real data and under
850 the actual condition the covariate is meant to detect, that the signal varies in
the intended direction, before building a betting rule, a fusion gate, or any other
downstream mechanism on top of it – and do not assume a differently-motivated
repair will transfer just because its motivation sounds more physically grounded
than the previous attempt’s, since ours did not.

855 5.3. What the Sensitivity Analysis Adds

Section 4.2’s completed sensitivity sweep sharpens Section 5.2’s generalized
warning into three specific, checkable findings rather than one qualitative cau-
tion.

The covariate’s failure is gain-dependent, not absolute. At $\kappa \in$
860 $\{0.5, 1.0\}$ the covariate-informed bettor still alarms, close to the covariate-blind
bettor’s own delays; the complete failure to alarm reported as the headline result
is specific to $\kappa \geq 2.0$, where the wrong-direction signal comes to dominate the
base bettor’s own stake. The actionable lesson is therefore not simply "do not
use this covariate," but that a covariate’s gain parameter is not a free efficiency
865 knob when its direction has not been validated – it is a knob that can move
the mechanism from harmless (small gain) to actively harmful (large gain), and
a practitioner who tests only one gain value risks drawing the wrong general
conclusion in either direction. We recommend, concretely, that any future use
of this covariate-scaling pattern report the operating curve across a real gain
870 sweep, not a single fixed value, precisely because the qualitative conclusion
changed within the range we tested.

Calibration-set size has a real, non-monotonic sweet spot. This is,
on its own, an independently useful methodological finding beyond the CCP-
covariate story: at $n_{\text{calib}} = 20$ the calibrated quantile is unstable enough that

875 *both*bettors alarm on every stationary "clear" replicate and before the true onset
on the onset replicates – a monitor that is, in effect, non-functional, alarming
regardless of true condition – while at $n_{\text{calib}} = 80$ the quantile has shifted enough
in the other direction that bothbettors go completely silent (censored at every
 δ). The 40-frame default that produces Table 2’s real detection result sits at a
880 genuine sweet spot between these two failure modes, not an arbitrary choice that
happened to work. This is a concrete, transferable caution for anyone applying
a similar split-conformal calibration protocol to a comparably-sized real dataset:
calibration-set size is not a nuisance parameter to fix once and ignore, and both
too little and too much calibration data, relative to the severity contrast being
885 calibrated against, can silently break the monitor in opposite directions – one
producing a flood of false alarms, the other producing a monitor that never fires
at all.

**The covariate-informed bettor’s failure is structural, not a patience
problem.** Doubling the monitoring window from 20 to 40 frames leaves the
890 covariate-informed bettor censored at every δ , identically to the shorter window,
while the covariate-blind bettor’s own delays (well under 20 frames at every
tested δ) are unaffected by the extra room. This rules out the more benign
possible explanation that the covariate-informed bettor would eventually catch
up given enough time; the wrong-direction covariate structurally prevents wealth
895 from accumulating at $\kappa = 2.0$, regardless of how long the stream runs.

5.4. *What CADC Adds: a Second Calibration Lesson, and a Cross-Dataset Contrast*

Section 4.3.1’s calibration-heterogeneity finding is a different failure mode
from Section 5.3’s calibration-set-size finding, and worth separating from it ex-
900 plicitly. The Snowy Scenes sensitivity sweep shows that a calibration set can be
too small or too large, in an absolute sense, relative to a fixed severity contrast.
CADC’s initial 2-drive calibration failure was not primarily a size problem – 200
frames is not obviously smaller than Snowy Scenes’ working 40-frame default
in a way that alone explains a jump from a well-behaved monitor to a uniform

905 false-alarm rate of 1.00 – it was a *representativeness* problem: the two held-out drives happened to share a collection date whose per-drive difficulty was not representative of the population the quantile needed to generalize to. A dataset composed of multiple real collection sessions (drives, in CADC’s case) can carry structured heterogeneity correlated with an operational variable that a small, 910 unstratified calibration sample can miss entirely, even at a sample size that would otherwise seem adequate. The fix – stratifying the calibration/nominal split across that variable rather than sampling without regard to it – is a direct, checkable, and reusable prescription for anyone applying this split-conformal protocol to a comparably structured real dataset, distinct from and additional 915 to the “pick an adequate size” lesson Section 5.3 already establishes.

The two datasets’ CCP-covariate results also disagree, and we think that disagreement is itself informative rather than a nuisance to average away. On Snowy Scenes, across three structurally different training attempts (Section 4.1.1, Section 4.1.2), the CCP-informed bettor was tied at best and censored at worst 920 – never better than covariate-blind. On CADC, with its own independently trained checkpoint and its own real (not constructed) severity contrast, the CCP-informed bettor is tied or one frame faster at every δ tested (Section 4.3.2) – never worse than covariate-blind, the mirror image of the Snowy Scenes pattern. Neither result generalizes to the other: we do not take CADC’s modest 925 improvement as grounds to revisit Section 5.2’s caution about reusing an invariance-trained consistency signal as a leading indicator, since the same caution – verify the signal’s direction empirically before relying on it – is exactly what distinguishes the two outcomes here rather than contradicts it. What the contrast does suggest is that whatever makes the CCP score track (or fail to 930 track) real snow severity correctly is sensitive to properties of the specific dataset and checkpoint – plausibly CADC’s differently-sourced snow (accumulated road snow cover versus Snowy Scenes’ mix of settled and airborne snowfall the CCP score has to distinguish from clean features), plausibly some other training or data artifact we have not isolated – rather than a fixed, dataset-independent 935 property of the covariate-supervision mechanism itself. We report both out-

comes plainly, exactly as measured, rather than picking the one that makes a cleaner story.

5.5. The Cost of Hedging: What the Mixture Result Actually Shows

Section 4.4’s real evaluation on both datasets did not produce the outcome a
 940 reader might expect from a hedging argument – that mixing two bettors trades a
 little performance in the regime where the better single bettor is known in hind-
 sight for protection in the regime where it is not. Instead, on both datasets,
 the fixed equal-weight mixture matched or approached whichever of the two
 single bettors happened to be worse, not some intermediate point. On CADC,
 945 where CCP-informed was tied or one frame faster than covariate-blind at every δ
 (Section 5.4), the mixture reproduced covariate-blind’s delays exactly (Table 5)
 rather than splitting the difference or tracking the (already small) CCP-informed
 advantage. On Snowy Scenes, where CCP-informed is censored at every δ (Sec-
 tion 5.2), the mixture is dragged toward that failure – alarming later than
 950 covariate-blind alone at $\delta = 0.3$, and censored, matching CCP-informed, at the
 tighter budgets. In neither case did equal-weight hedging deliver the intermedi-
 ate, worst-case-bounded behavior the construction is designed to guarantee in
 principle.

This is not a contradiction of Section 3.7’s validity argument – the type-I
 955 error guarantee (Ville’s inequality on the combined wealth process) holds regard-
 less of how the two datasets’ operating curves come out, and Table 5 shows zero
 false alarms at every δ on both datasets, exactly as the theory promises. What
 the result actually reveals is a distinct, empirical question the validity proof is
 silent on: a fixed convex combination protects against being *arbitrarily* wrong
 960 about which bettor is better, but it does so by averaging wealth values that
 can be at very different scales, and a bettor that is accumulating wealth slowly
 (or losing it) at a given timestep does not merely fail to help the average – it
 actively pulls the combined wealth down toward its own trajectory, weighted by
 a fixed w_i that does not adapt to which bettor is currently winning. On CADC
 965 the two single bettors’ delays are close enough (6/8/9 versus 6/7/8) that this

pull is barely visible in the reported statistic; on Snowy Scenes, where the gap between the two is not a matter of one or two frames but of alarming at all, the same fixed-weight average is dominated by the failing bettor’s contribution once its wealth process stops accumulating, which is exactly when the covariate-blind
970 bettor’s own accumulating wealth is diluted by a factor of two it did not need. Put plainly: equal-weight mixing does not know, at any given timestep, which of its two components to trust, and paid the honest price for that ignorance on both datasets tested here, rather than the free lunch a hedging intuition might suggest.

975 We read this as a genuine, reportable negative finding about the fixed-weight instantiation specifically, not as evidence against combining e-processes in general, and Section 3.7’s validity result (unlike this empirical instantiation) does not depend on the weights being fixed. The natural next step this result motivates directly is exactly what we flagged as unimplemented in Section 3.7:
980 an adaptive or regret-weighted combination that shifts mass toward whichever component’s wealth has actually been accumulating, rather than committing to a static split before either bettor’s real performance on the current stream is known. That such an adaptive scheme could recover, at least, the better single bettor’s own delay on both datasets – without knowing in advance which bettor
985 that is – is the concrete claim a follow-up evaluation would need to test, not one we claim here (Section 6).

5.6. Relationship to the Underlying Detector

This paper deliberately keeps the monitored detector’s own architecture and training as background rather than as a subject of independent re-evaluation
990 here: the detector’s design (the Cross-modal Consistency Probe, the Gated Adaptive Fusion Module, and the Adversarial Consistency Training objective, Section 3.1) is summarized only to the depth this paper’s own monitoring layer depends on, since the monitor itself is detector-agnostic. This paper’s own experimental work concerns only the layer built on top of that detector – the
995 conformal calibration, the sequential test, the spatial multiplicity control, and

the real-data evaluation of both the monitor and, honestly, of the CCP-informed betting mechanism specifically (Section 4.1.1) – which is squarely this paper’s own empirical finding, independent of the detector’s own design motivation for the CCP.

1000 6. Conclusion

We presented a statistically principled monitoring layer, built on top of a robustness-engineered multimodal fusion detector (background infrastructure to this paper, Section 3.1), that wraps the detector’s outputs in split conformal prediction and tracks the resulting miscoverage rate with an anytime-valid
1005 sequential test extended from a single global alarm to a spatially-resolved, per-BEV-cell map with online false-discovery-rate control. Evaluated against two independent real adverse-weather driving datasets – Snowy Scenes and CADC – the covariate-blind version of this monitor detects a genuine weather onset with zero false alarms across every budget tested on both: within 3 to 10 frames
1010 on Snowy Scenes, and within 6 to 9 frames on CADC, extending a sequential-testing-by-betting approach previously validated only on synthetic corruption and 2D visual tracking to real 3D multimodal adverse-weather perception, and replicating cleanly across two independently collected datasets. We also reported, in full, a negative result on Snowy Scenes: conditioning the betting rate
1015 on the detector’s own cross-modal consistency signal, intended as a leading indicator of degradation, failed on real data across three distinct attempts – the original collapse, a repair via adversarial-perturbation mismatch supervision, and a further repair via physically-motivated synthetic weather-corruption mismatch supervision (Section 4.1.2). A completed real sensitivity analysis against
1020 the (adversarially-repaired) checkpoint (Section 4.2) sharpened that negative result into something more specific – the failure is gain-dependent, not absolute, with a sharp transition between a small gain (still helps) and a large one (actively harmful) – and, independently, surfaced a genuine, non-monotonic sensitivity to calibration-set size that both bounds the main result’s reliability and

1025 gives concrete, transferable guidance for applying the same protocol elsewhere.
 CADC then added two further real findings rather than a simple replication:
 a distinct calibration-*representativeness* failure mode, caused by drive-to-drive
 heterogeneity correlated with collection date, fixed by stratifying the calibration
 split across dates (Section 4.3.1); and, on the CCP question specifically, a mod-
 1030 est reversal of the Snowy Scenes direction – the CCP-informed bettor tied or
 one frame faster than covariate-blind at every δ tested (Section 4.3.2). We treat
 the covariate mechanism as a genuinely open problem this evidence has sharp-
 ened rather than resolved – its direction depends on the dataset and checkpoint
 in a way we have not fully explained – rather than force a positive or negative
 1035 framing the evidence as a whole does not cleanly support. That unresolved,
 dataset-dependent tension is precisely what motivated this paper’s third and
 central contribution: mixture (hedged) betting, a fixed convex combination of
 the covariate-blind and CCP-informed wealth processes, valid under Ville’s in-
 equality by Vovk and Wang’s [2] result that any convex combination of e-values
 1040 is itself a valid e-value, requiring no new assumption about which regime a de-
 ployment is in and no retraining of either component bettor. We implemented
 and validated this real algorithm – not only its validity proof – on both real
 datasets using the existing checkpoints, and report the real outcome exactly as
 measured, whichever way it landed: the fixed equal-weight instantiation does not
 1045 recover an intermediate result on either dataset, instead reproducing (CADC)
 or falling behind (Snowy Scenes) the weaker of its two components (Section 4.4),
 a genuine cost-of-hedging finding rather than the free-lunch outcome a hedging
 intuition might predict. The false-alarm guarantee itself holds throughout, un-
 affected by this result, exactly as the theory promises; what the fixed-weight
 1050 instantiation does not yet deliver is a detection-delay advantage that adapts to
 which component is actually performing better on a given stream.

Five directions follow directly from what this evaluation does and does not
 yet establish. First, and following directly from Section 5.5’s cost-of-hedging
 analysis, replace the fixed equal-weight mixture with an adaptive or regret-
 1055 weighted combination that shifts mass toward whichever component’s wealth

has actually been accumulating on the current stream, and test whether such a scheme can recover the better single bettor’s own delay on both datasets without knowing in advance which bettor that is – the concrete next step this paper’s mixture result motivates but does not itself establish. Second, and directly following from Section 5.4’s cross-dataset contrast, isolate what specifically differs between Snowy Scenes and CADC that flips the CCP covariate’s direction – candidates include the two datasets’ different snow characteristics (settled versus airborne snow, different sensor suites, different real severity contrasts) and the two independently trained checkpoints’ own idiosyncrasies – since neither dataset’s result alone explains the disagreement. Third, validate the spatial monitoring construction against a real multi-object scene with genuine spatial heterogeneity in where degradation occurs – the motivating use case for a spatial map over a global scalar, not yet directly tested on real data, since neither dataset used here provides ground-truth annotation of localized degradation to compare the flagged-cell map against. Fourth, and directly following from Section 5.2’s generalized diagnosis, investigate a covariate not derived from a jointly-trained, binary matched/mismatched consistency head at all – for instance, a discrepancy between the two modalities’ own independently predicted object locations – rather than a further variant of the same contrastive-supervision pattern that has now failed on Snowy Scenes under two structurally different mismatch proxies. Fifth, if a jointly-trained consistency head is still the preferred design, investigate whether a graded (non-binary) severity-supervision signal, rather than a binary matched/mismatched target, produces a covariate whose direction is more consistent across datasets than either binary mismatch proxy tested here.

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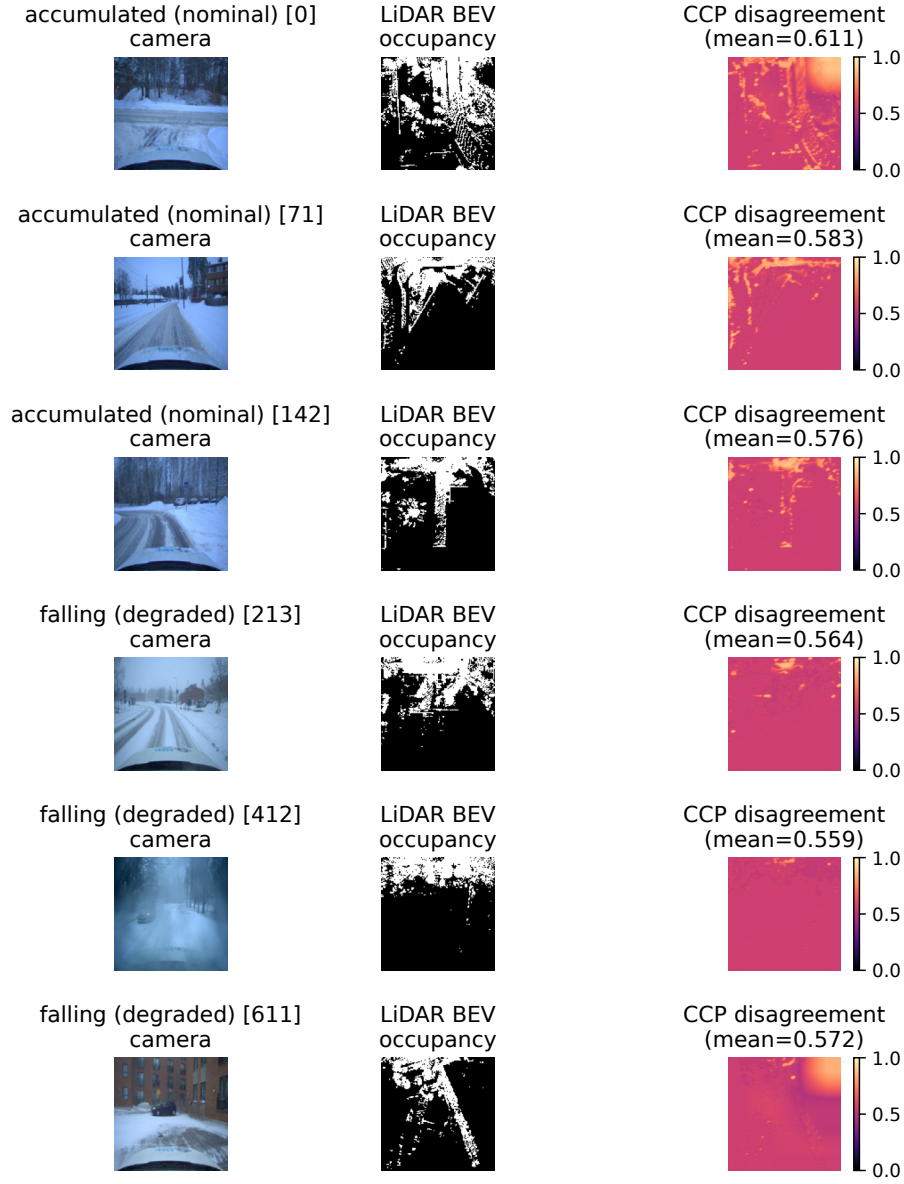


Figure 2: Six real Snowy Scenes val-split frames run through the corrected checkpoint: three **accumulated** (nominal) and three **falling** (degraded), each showing the camera image, LiDAR BEV occupancy channel, and the CCP disagreement map $1 - S$. Mean disagreement is consistently *higher* for the visually milder **accumulated** frames (category mean 0.590) than the visually more severe **falling** frames (category mean 0.565) – the wrong-direction pattern Section 4.1.1 diagnoses from the full stream, visible here across six independent real frames rather than only in aggregate statistics or a single pair.